

Yash Jayswal

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ML Systems & Inference Optimization | ASR · LLM Serving · Distributed Training

✉ yashmjayswal@gmail.com ☎ +91 9909983123 📁 Portfolio 🌐 GitHub in Yash Jayswal

Education

INDIAN INSTITUTE OF TECHNOLOGY, DELHI (IITD)

Delhi, India

B.Tech. in Electrical Engineering

Nov 2020 – Aug 2024

- **Coursework:** Probability & stochastic processes, Digital image processing, Machine intelligence & learning, Graph theory.

Experience

MLSys Group, UC SAN DIEGO

remote

Research Assistant

June 2026 – Sept 2026

- Built **C++ indexing** and **inference optimization** for **embedding** and **reranker serving** in CodeNib, a **multi-view repository context** system for **coding agents** (see Publications).
- Improved **token efficiency** and reduced **retrieval latency** for **LLM-based AI agents** via **low-latency data systems**.

SAMSUNG HQ

Suwon, South Korea

ML/AI Speech Engineer

Aug 2024 – ongoing

- Fine-tuned a Qwen MoE streaming ASR model using LoRA adapters; investigated WER-latency tradeoffs under streaming constraints, achieving 8% WER on multilingual transcription.
- Developed a multilingual ASR model (31 languages) and optimized it for real-time streaming using quantization, chunking, K-V caching, VAD, and align-attention, achieving $\approx 45\%$ latency reduction from baseline.
- Built **C++ TensorRT** inference pipelines and **Python ONNX Runtime** ASR inference code, including **ONNX-to-TensorRT** conversion workflows for low-latency deployment.
- Implemented scalable C++ model handling modules within a multi-user Bixby Cloud environment to enable deployment of inference-optimized Fast Conformer-based ASR models (both encoder-decoder and LSTM architectures).
- Designed an end-to-end robust Data collection pipeline for low-latency training ASR models' training.

SAMSUNG HQ

Suwon, South Korea

NLP Intern

May 2023 – July 2023

- Developed **Name Entity Replacer** for the Personal Database module with tolerance toward mispronounced names.
- Proposed novel phoneme similarity metric based on manner of articulation paired with Metaphone and Soundex algorithm.
- Developed Metaphone and phoneme similarity metrics, achieving **96.3%** and **98.3%** accuracy respectively for spoken name matching, reducing recognition errors by $\approx 55\%$.

Research & Systems Projects

LLM Scaling & Systems Optimization (JAX/TPU) | [Workflow](#) [🔗](#)

[GitHub](#) [🔗](#)

- Built distributed **Transformer MLPs** in **JAX** evaluating **TP**, **PP**, and **FSDP**. Hand-coded a **Mixture of Experts (MoE)** layer via **jax.shard_map** and **All-to-All** routing to bypass $O(N \times S \times D)$ **AllGather** memory bottlenecks.
- Leveraged **XLA trace profiling** on a **TPU v5e-8 cluster** to hide communication latency, achieving a **43x MoE speedup** (**73ms** $\rightarrow$ **1.7ms**) and optimizing it using **ReduceScatter** operation for matmuls to $\approx 1.1\mu s$.

Vision Semantic Communication | [Prof. Brejesh Lal](#) [🔗](#)

[report](#) [🔗](#)

- Developed a **generative AI-based** semantic communication pipeline for image communication targeting **6G** communication.
- Proposed a **guided Stable Diffusion + ControlNet** approach using image captions, edge maps, and scaled-down latent representations, achieving reduced hallucination, higher noise tolerance, and improved color retention under high compression.
- Achieved a 0.31 LPIPS score on VGG net in noisy communication channel, surpassing the previous best score by **40%**.

Publications

A Simple Model for Panic Driven Motion in a Refugee Camp | [JHBSE 2026](#) [🔗](#)

July 2026

Simulated crowd dynamics via numerical mollification and RK4; optimized obstacle configurations reducing fatalities by **70%**.

CodeNib: Multi-View Data System for Coding Agents | [arXiv 2026](#) [🔗](#)

July 2026

Built incremental multi-view repository context system; with 8.7x/25.4x faster updates and 50–87% fewer agent trajectory tokens.

Technologies

Software Skills: Python, C, C++, Tensorflow, Jax, Numpy, PyTorch, Kubernetes, TensorRT, CUDA, Docker, Algorithms

AI Skills: Low latency AI Inference, Deep Learning, Machine Learning, Simulation, LLM finetuning, Distributed Training, Data curation, ASR, Vision, LLM inference

Scholastic Achievements

- **Joint Entrance Examination Advanced:** Secured **All India Rank 319** among 150,000+ students
- **Summer Undergrad Research Award (SURA):** Conferred by Industrial R&D Unit, IIT Delhi for excellent research